

Gaia DR3 Colour–Magnitude Diagram

A Quality-Controlled Analysis of Local Stellar Populations from Gaia DR3 Astrometry and Photometry

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Barcelona, Spain · August 2026

Abstract

This report presents a reproducible analysis of a nearby Gaia Data Release 3 (DR3) stellar sample, built into a quality-controlled colour–magnitude diagram (CMD). Starting from a fixed 5,000-source Gaia archive snapshot, a photometric-quality cut on the colour-corrected BP/RP flux-excess factor retains 4,681 sources. The resulting CMD is used to identify white-dwarf and high-luminosity candidate regions and to model the observed main-sequence ridge with a third-order empirical polynomial (RMSE = 0.120 mag). A residual and sensitivity analysis shows that parallax precision is a material contributor to the observed faint-side scatter around the ridge: tightening the parallax signal-to-noise requirement from 5 to 10 reduces faint-side residual outliers from 157 to 20. The analysis is presented together with its assumptions and limitations, including the use of direct parallax inversion and the absence of an extinction correction.

1. Introduction and Objective

A colour–magnitude diagram (CMD) is the observational counterpart of the Hertzsprung–Russell diagram: it plots a star's colour, here Gaia BP–RP, against its absolute magnitude, here M_G in the Gaia G band. Because colour tracks surface temperature and absolute magnitude tracks intrinsic luminosity, the CMD separates stellar populations — the main sequence, evolved high-luminosity stars, and compact white dwarfs — into recognisable regions.

The objective of this work is to test whether Gaia DR3 astrometry and photometry, filtered with straightforward, reproducible quality criteria, can be used to (i) build a clean CMD from a nearby stellar sample, (ii) identify broad stellar-population regions within it, and (iii) quantitatively describe the main sequence and the scatter of individual stars around it.

The analysis is deliberately conservative in scope. Two assumptions bound its validity: distances are obtained by direct parallax inversion, which is only reliable at the parallax precision required in the sample selection, and no correction for interstellar extinction or reddening is applied. Both choices are discussed further in Section 6.

2. Gaia Data and Quality Control

2.1 Sample selection

The input sample is a fixed snapshot of 5,000 sources drawn from the Gaia DR3 `gaiadr3.gai_source` table via the Gaia Archive. The ADQL query applied nearby-sample and basic astrometric-quality criteria at retrieval time:

- `parallax > 2 mas` (approximately within 500 pc);
- `parallax_over_error > 5`, ensuring a usable parallax signal-to-noise ratio;

- $ruwe < 1.4$, a standard astrometric-solution quality cut;
- $visibility_periods_used \geq 8$, requiring adequate observational coverage;
- non-null G, BP and RP mean magnitudes.

Because an unordered TOP 5000 query is not guaranteed to return the same rows on a later run, the query is retained in the analysis notebook only for provenance. The reproducible input for this report is the stored ECSV snapshot, so re-running the notebook reproduces the results reported here exactly.

2.2 Photometric quality cut

Gaia's `phot_bp_rp_excess_factor` flags sources whose BP/RP photometry is inconsistent with the G-band flux, but its expected value depends on colour. The colour-corrected flux-excess factor, C^* , removes this dependence via the standard Gaia DR3 piecewise correction. Sources are retained only if:

$$-0.08 < C^* < 0.20$$

This boundary is a documented project choice rather than a universal Gaia standard. Applying it to the 5,000-source initial sample leaves 4,681 quality-controlled sources (93.6% retention), which form the working sample for the rest of the analysis.

3. Derived Quantities and the Colour-Magnitude Diagram

For each retained source, distance is obtained by direct parallax inversion, and this distance is combined with the apparent G magnitude to give the absolute magnitude:

$$d \text{ (pc)} = 1000 / \varpi \text{ (mas)}$$

$$BP - RP = m_{BP} - m_{RP}$$

$$M_G = m_G - 5 \log_{10}(d) + 5$$

All 4,681 sources yield finite, positive distances and finite CMD coordinates, consistent with the `parallax_over_error > 5` requirement imposed at the query stage.

Two complementary views of the resulting CMD are produced: an individual-source scatter plot, which preserves outliers and sparse regions, and a two-dimensional density histogram, which highlights where the population is concentrated. In the magnitude axis is inverted throughout, following astronomical convention, so intrinsically brighter stars appear towards the top of each figure.

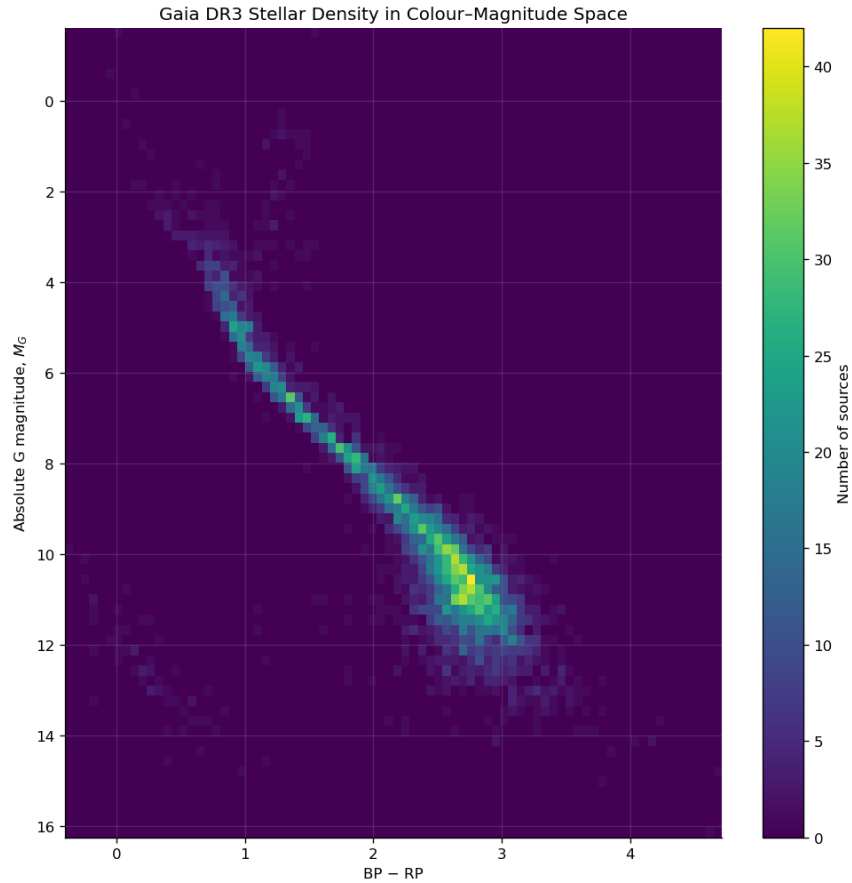


Figure 1. Density colour-magnitude diagram of the 4,681 quality-controlled Gaia DR3 sources, binned in BP-RP and M_G . The dominant main-sequence locus is clearly resolved, with the highest source density around BP-RP $\approx$ 2.6–2.8.

4. Population and Main-Sequence Results

4.1 Candidate population regions

Three broad, non-exclusive working regions are defined directly on the CMD:

- White-dwarf candidates: $M_G > 6 + 5(BP-RP)$ and $BP-RP < 1.5$ — 59 sources (1.26% of the working sample);
- High-luminosity candidates: $M_G < 4$ — 189 sources (4.04%);
- Main-sequence working region: $0.5 < BP-RP < 3.2$ and $4 < M_G < 12.5$ — 4,211 sources (89.96%).

These regions are deliberately simple linear or rectangular cuts, chosen for reproducibility rather than completeness. They identify candidates only: no independent spectroscopic or evolutionary confirmation is applied, so individual sources — particularly close to a boundary — may be misclassified.

4.2 Main-sequence ridge model

Within the main-sequence working region, the median M_G is computed in 39 colour bins, and a third-order polynomial is fitted to this median ridge as an empirical description of the sample's central main-sequence locus. The fit achieves an RMSE of 0.120 mag relative to the binned medians, indicating that a low-order polynomial captures the ridge shape well over the fitted colour range. The fitted ridge, together with the white-dwarf and high-luminosity candidates, is shown in Figure 2.

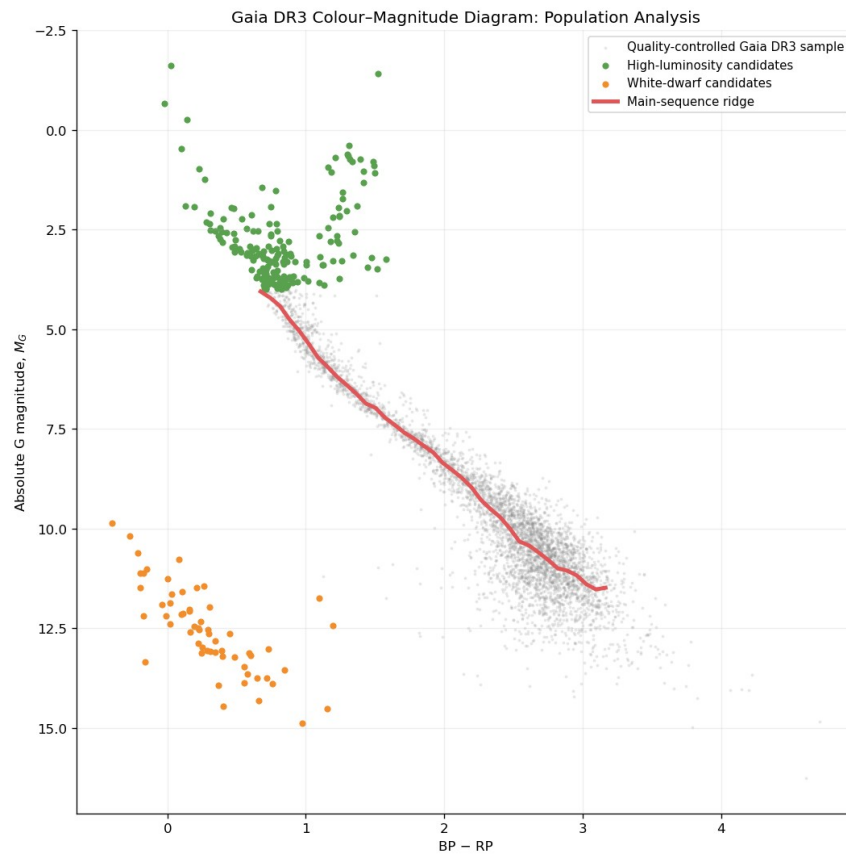


Figure 2. Final annotated Gaia DR3 colour-magnitude diagram, showing the quality-controlled sample (grey), high-luminosity candidates (green), white-dwarf candidates (orange), and the fitted third-order main-sequence ridge (red).

The polynomial is an empirical description of this specific sample and should not be interpreted as a general stellar-evolution relation; a different Gaia selection, distance range, or metallicity mix would in general produce a different ridge.

5. Robustness Analysis

For each main-sequence source, the residual relative to the fitted ridge is $\Delta M_G = M_{G,\text{observed}} - M_{G,\text{fit}}$. Sources with $|\Delta M_G| > 1.5$ mag are flagged as strong residual outliers for diagnostic purposes; they are not removed from the sample.

The robust scatter of individual stars around the ridge is approximately 0.491 mag — substantially larger than the 0.120 mag RMSE of the median ridge itself, confirming that individual-star photometric and physical scatter (e.g. unresolved binarity, metallicity spread, residual distance error) dominates over any imperfection in the ridge fit. Of the 4,211 main-sequence sources, 157 are flagged as strong residual outliers, and inspection shows these concentrate on the faint side of the ridge.

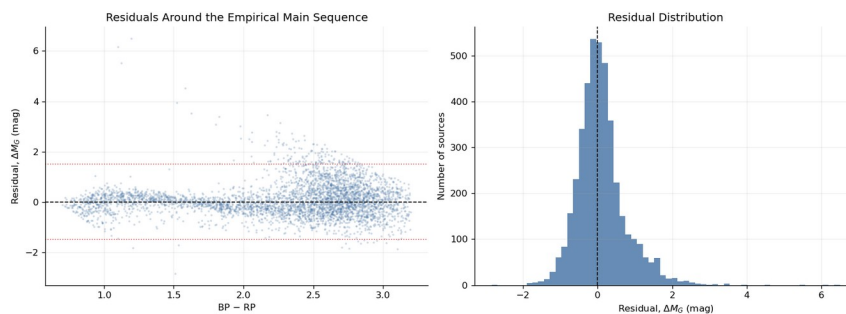


Figure 3. Left: individual main-sequence residuals ΔM_G as a function of BP–RP, with the ± 1.5 mag outlier threshold marked. Right: the corresponding residual distribution, showing a peaked core with an extended faint-side (positive ΔM_G) tail.

To test whether this faint-side tail is driven by distance (parallax) precision rather than intrinsic physical scatter, the parallax signal-to-noise requirement is tightened from $S/N > 5$ (the original query-level cut) to $S/N > 10$, and the faint-side outlier count is recomputed under the stricter cut.

Result	Value
Faint-side residual outliers, parallax $S/N > 5$	157
Faint-side residual outliers, parallax $S/N > 10$	20
Reduction	$\approx 87\%$

Restricting to higher-precision parallaxes removes the large majority of faint-side outliers, showing that parallax precision is a material contributor to the apparent faint-side broadening of the main sequence. The remaining 20 outliers indicate that parallax precision does not explain every residual, and that some additional scatter — plausibly photometric noise, unresolved multiplicity, or genuine population spread — persists even at higher astrometric precision.

6. Conclusions and Limitations

6.1 Summary of results

Result	Value
Initial Gaia DR3 snapshot	5,000
Final quality-controlled sample	4,681
White-dwarf candidates	59 (1.26%)
High-luminosity candidates	189 (4.04%)
Main-sequence working sample	4,211 (89.96%)
Third-order ridge-fit RMSE	0.120 mag
Robust observed scatter around ridge	0.491 mag
Faint-side outliers (parallax $S/N > 5$)	157
Faint-side outliers (parallax $S/N > 10$)	20

Starting from 5,000 nearby Gaia DR3 sources already meeting query-level astrometric criteria, a colour-corrected BP/RP flux-excess cut produced a final sample of 4,681 sources. The resulting CMD shows the expected dominant main sequence, a bright high-luminosity region, and a separate faint-blue region consistent with white dwarfs. The main-sequence working region, containing 4,211 sources, is well described by a third-order empirical ridge (RMSE ≈ 0.120 mag), while individual stars scatter substantially more broadly around it (≈ 0.491 mag). This scatter is concentrated on the faint side of the ridge, and a sensitivity test shows that parallax precision is an important, though not sole, contributor to that broadening.

6.2 Limitations

- Direct parallax inversion ($d = 1000/\varpi$) is used because the sample already satisfies parallax $S/N > 5$; it is not a general-purpose distance estimator and would be inappropriate for lower-precision parallaxes.
- No interstellar extinction or reddening correction is applied, so BP–RP and M_G are observed rather than intrinsic quantities. This is a plausible contributor to the residual scatter that is not addressed here.
- The white-dwarf and high-luminosity regions are simple, reproducible CMD cuts, not confirmed stellar classifications; individual sources near a boundary may be misclassified.
- The main-sequence polynomial is an empirical description of this specific 4,681-source sample and should not be treated as a universal stellar-evolution relation.
- The input is a fixed, nearby 5,000-source snapshot, so the reported population fractions describe this sample only and are not Milky Way–wide estimates.

Overall, the analysis shows that a small, well-documented set of Gaia DR3 quality cuts is sufficient to recover a physically sensible CMD and to quantify, rather than merely observe, its main-sequence scatter — while making clear which parts of that scatter are explained by astrometric precision and which remain open.